

Rajdeep Gill

(431) 998-9785 | hi.rajdeepgill@gmail.com | github.com/Rajdeep-Gill

EDUCATION

Bachelor of Science in Computer Engineering (Co-op)

University of Manitoba

May 2026

Winnipeg, MB

GPA: 4.45/4.5

Relevant Coursework: *Data Structures and Algorithms, Systems Engineering Principles, Parallel Programming, Digital System Design, Microprocessor Interfacing*

EXPERIENCE

Robinhood

Software Engineer Intern

May 2025 - Sept 2025

Toronto, ON

- Incoming intern for the summer 2025 term, working on the brokerage team.

Biosensors Research Lab - University of Manitoba

Undergraduate Research Student

May 2024 - Dec 2024

Winnipeg, MB

- Designed and trained a convolutional neural network to predict biological cell viability from 10,000+ diffraction pattern images, achieving **85%** accuracy and demonstrating potential for non-invasive cell viability assessment
- Enhanced particle diffraction centroid detection using computer vision, improving success rate by **17%** (from 75% to 92%) and providing more accurate data for the CNN model
- Developed a Python-based data visualization tool with an interactive GUI to process, filter, and display diffraction patterns and model predictions, enabling efficient data analysis and interpretation
- Utilized unsupervised machine learning techniques to help group and classify cells based on impedance measurements, providing insights into cell health and viability

Biosensors Research Lab - University of Manitoba

Undergraduate Research Student

May 2023 - Aug 2023

Winnipeg, MB

- Applied dimensionality reduction techniques (t-SNE, PCA) to analyze and cluster datasets, uncovering key correlations for future research directions
- Optimized data analysis workflows by implementing parallel processing, reducing processing time by 90%
- Developed an oscilloscope interfacing program to automate data collection, improving speed and accuracy

PROJECTS

Google Docs Clone | *React, TypeScript, HTML/CSS, Next.js*

- Developed a Google Docs clone with real-time collaboration, user authentication, and document sharing
- Utilized Convex to store and sync document data, allowing multiple users to edit in real-time
- Implemented organization-wide document sharing and user permissions to control access to documents

Financial Dashboard | *React, TypeScript, HTML/CSS, Next.js, Hono.js*

- Created an expense and income tracking dashboard with user authentication and real-time data visualization
- Implemented a RESTful API to handle user data and transactions, using a PostgreSQL database to store user information

Deaf Curling Accessibility Aid | *C, ESP-32, Fusion360*

- Developed a device to assist deaf curlers in receiving instructions from their skip during games
- Designed a PCB and 3D-printed enclosure ensuring durability and portability

PUBLICATIONS

B. Arzhang, E. Kovacs, J. Lee, **R. Gill**, E. Salimi, D. Thomson, and G. Bridges, "Machine Learning Classification of Microplastics by Integrating Optical and Dielectrophoresis Features," in *IEEE AP-S/URSI 2025*.

B. Arzhang, G. Bridges, **R. Gill**, J. Lee, E. Kovacs, E. Salimi, and D. Thomson, "Dielectric Analysis of Single Biological Cells at Both Microwave and Optical Frequencies," in *IEEE AP-S/URSI 2025*.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML/CSS, R, SQL, MATLAB

Frameworks & Libraries: React, Next.js, TensorFlow, PyTorch, OpenCV,

Developer Tools: Git, VSCode, Visual Studio, Android Studio, IntelliJ